LA-UR- 12-00353

Approved for public release; distribution is unlimited.

Title:	Structural Health Monitoring of Wind Turbine Blades under Fatigue Loads
Author(s):	Sanuel Dyas, Colorado School of Mines Justin Scheidler, Ohio State University Stuart Taylor, INST-OFF, LANL Gyuhae Park, INST-OFF, LANL Kevin Farinholt, AET-1, LANL
Intended for:	IMAC XXX A CONFERENCE AND EXPOSITION ON STRUCTURAL DYNAMICS, Jacksonville, FL, January 30-February 2, 2012.

Los Alamos
NATIONAL LABORATORY
EST. 1943

Los Alamos National Laboratory, an affirmative action/equal opportunity employer, is operated by the Los Alamos National Security, LLC for the National Nuclear Security Administration of the U.S. Department of Energy under contract DE-AC52-06NA25396. By acceptance of this article, the publisher recognizes that the U.S. Government retains a nonexclusive, royalty-free license to publish or reproduce the published form of this contribution, or to allow others to do so, for U.S. Government purposes. Los Alamos National Laboratory requests that the publisher identify this article as work performed under the auspices of the U.S. Department of Energy. Los Alamos National Laboratory strongly supports academic freedom and a researcher's right to publish; as an institution, however, the Laboratory does not endorse the viewpoint of a publication or guarantee its technical correctness.

Form 836 (7/06)

STRUCTURAL HEALTH MONITORING OF WIND TURBINE BLADES UNDER FATIGUE LOADS

Samuel J. Dyas[1], Justin Scheidler[2], Stuart G. Taylor[3], Kevin Farinholt[3], Gyuhae Park[3]

[1] Dept. of Mechanical Eng., Colorado School of Mines, Golden, CO 80401
[2] Dept. of Mechanical Eng., Ohio State University, Columbus, OH 43210
[3] The Engineering Institute, Los Alamos National Laboratory, Los Alamos, NM 87545

Abstract

This paper presents the results of dynamic characterization and preparation of a full-scale fatigue test of a 9m CX-100 blade. Sensors and actuators utilized include accelerometers and piezoelectric sensors. To dynamically characterize a 9m CX-100 blade, full scale modal analyses were completed with varying boundary conditions and blade orientations. Also, multi-scale sensing damage detection techniques were explored; high frequency active-sensing was used in identifying fatigue damage initiation, while low frequency passive-sensing was used in assessing damage progression. Ultimately, high and low frequency response functions, wave propagations, and sensor diagnostic methods were utilized to monitor and analyze the condition of the wind turbine blade under fatigue loading.

Los Alamos NATIONAL LABORATORY EST. 1943

Structural Health Monitoring of a Wind Turbine Blade under Fatigue Loads

Conference Presentation: *IMAC 2012*

Samuel Dyas
-Colorado School of Mines

Justin Scheidler
-Ohio State Univ.



Stuart Taylor, Kevin Farinholt, Gyuhae Park
- The Engineering Institute, Los Alamos National Laboratory

Los Alamos NATIONAL LABORATORY EST. 1943

Outline

- Background
- Objectives
- Modal Analysis
- Accelerometer vs. Piezoelectric Sensor
- Structural Health Monitoring
- Conclusions





Los Alamos NATIONAL LABORATORY EST. 1943

Outline

- Background
- Objectives
- Modal Analysis
- Accelerometer vs. Piezoelectric Sensor
- Structural Health Monitoring
- Conclusions

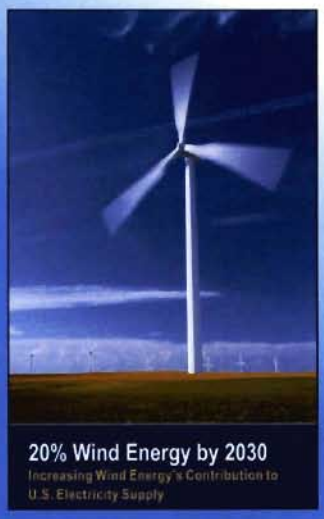




Los Alamos NATIONAL LABORATORY EST. 1943

Background

- Wind Energy
  - Fastest growing renewable domestic energy source in United States
- Structural Health Monitoring
  - Life based vs. Time based Maintenance
  - Blade damage can cause secondary damage



Los Alamos NATIONAL LABORATORY EST. 1943

Outline

- Background
- Objectives
- Modal Analysis
- Accelerometer vs. Piezoelectric Sensor
- Structural Health Monitoring
- Conclusions





Los Alamos NATIONAL LABORATORY EST. 1943

Objectives

1. Dynamically Characterize CX-100 in variety of boundary conditions







Los Alamos NATIONAL LABORATORY EST. 1943

Objectives

2. Prepare working and validated tools/methods for SHM of CX-100 during full scale fatigue test



[courtesy Sandia National Laboratory]

Los Alamos NATIONAL LABORATORY EST. 1943

Outline

- Background
- Objectives
- **Modal Analysis**
- Accelerometer vs. Piezoelectric Sensor
- Structural Health Monitoring
- Conclusions



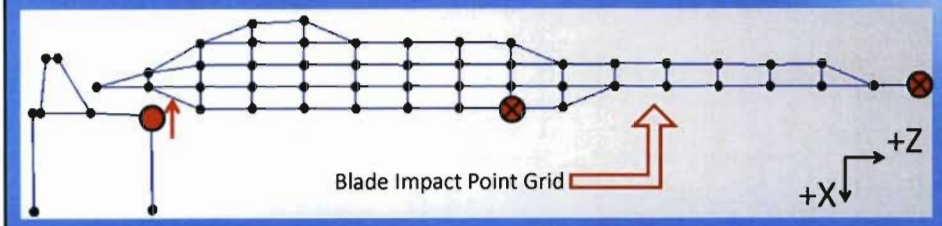


Los Alamos NATIONAL LABORATORY EST. 1943

Fixed-Free Modal Analysis



- Impact grid of 73 points
  - Blade: 48 in Y-direction, 1 in X-direction
  - Fixture: 14 in Y-direction, 10 in X-direction
- 3 shear accelerometers
- Impact hammer tip excited up to 150 Hz

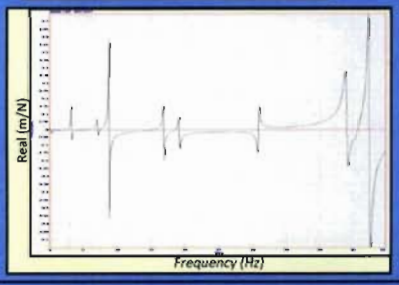


Los Alamos NATIONAL LABORATORY EST. 1943

Fixed-Free Results



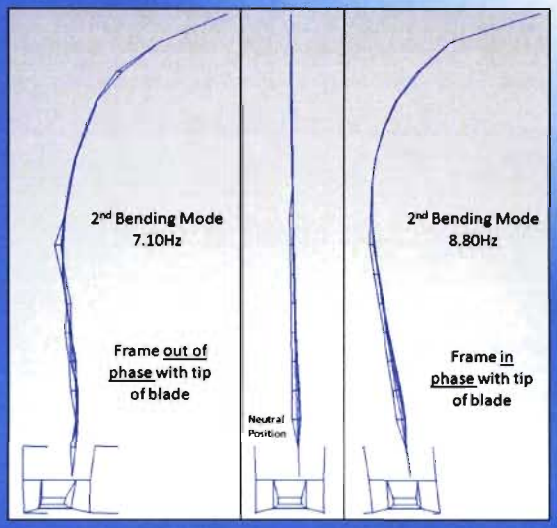
Mode	Frequency (Hz)	Damping (%)	Description
1	3.22	0.241	1st Flap Bending
2	4.16	0.251	1st Lag Bending
3	7.10	0.361	2nd Flap Bending
4	8.80	0.275	2nd Flap Bending
5	16.9	0.325	3rd Flap Bending
6	19.2	0.330	2nd Lag Bending
7	30.9	0.255	4th Flap Bending
8	39.3	0.565	1st Torsion
9	47.5	0.291	5th Flap Bending



Split Mode => Possible Tuned Absorber?

Los Alamos NATIONAL LABORATORY EST 1943

Split 2nd Bending Mode

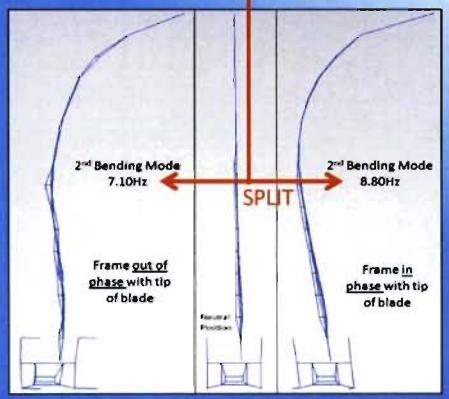


Los Alamos NATIONAL LABORATORY EST 1943

Fixture Analysis



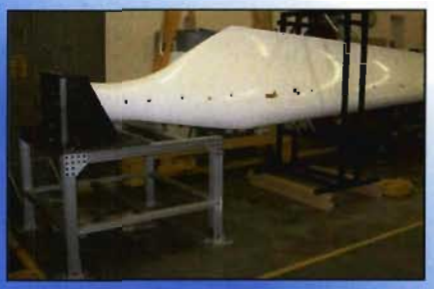
Mode	Frequency (Hz)	Damping (%)	Description
1	8.37	0.409	Side-to-Side Oscillation
2	27.1	0.388	Side-to-Side Torsion
3	45.7	0.555	Vertical Oscillation



Los Alamos NATIONAL LABORATORY EST. 1943

Free-Free vs. Fixed-Free



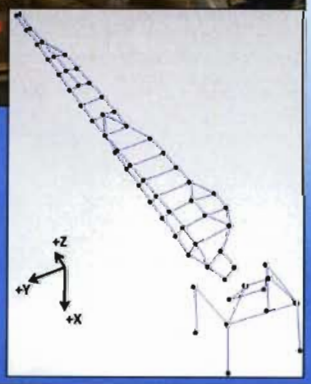


Mode	Descritpion	Free-Free Frequency (Hz)	Fixed-Free Frequency (Hz)	% Difference
1	1st Flap Bending	7.61	3.22	97.0%
2	1st Lag Bending	18.1	4.16	206%
3	2nd Flap Bending	20.2	8.8	93.0%
4	3rd Flap Bending	32.2	16.9	69.0%
5	2nd Lag Bending	45.1	19.2	96.2%
6	4th Flap Bending	50.5	30.9	51.1%
7	1st Torsion	63.9	39.3	50.5%

Los Alamos NATIONAL LABORATORY EST. 1943

Rotation of Blade





- Impact grid of 56 points
  - Blade: 30 X-direction, 18 Y-direction
  - Fixture: 8 in both X and Y direction
- 3 shear accelerometers
- Impact hammer tip excited up to 150 Hz

Los Alamos NATIONAL LABORATORY EST. 1943

Horizontal vs. Vertical Fixed-Free





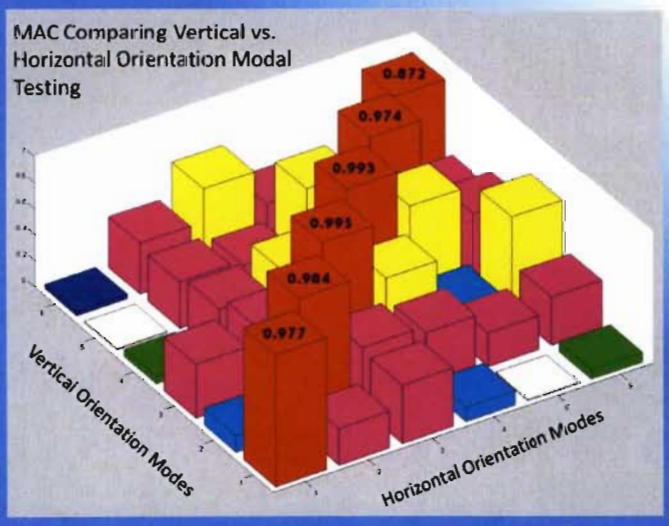
Mode	Description	Vertical Orientation		Horizontal Orientation		Percent Difference	
		Frequency (Hz)	Damping (%)	Frequency (Hz)	Damping (%)	Frequency (Hz)	Damping (%)
1	1st Flap Bending	3.22	0.24	3.37	0.29	4.55%	19.0%
2	2nd Flap Bending	8.80	0.28	8.77	0.24	0.34%	15.5%
3	3rd Flap Bending	16.9	0.33	16.8	0.31	0.59%	6.26%
4	4th Flap Bending	30.9	0.26	31.0	0.31	0.32%	17.7%
5	1st Torsion	39.3	0.57	38.8	0.42	1.28%	31.0%
6	5th Flap Bending	47.5	0.29	52.3	1.14	9.64%	184%

-UNCLASSIFIED-

2011 Los Alamos Dynamic Summer School
S. Dyas & J. Scheidler

Los Alamos NATIONAL LABORATORY EST. 1943

Horizontal vs. Vertical Fixed-Free



-UNCLASSIFIED-

Los Alamos NATIONAL LABORATORY EST. 1943

NREL Modal Analysis







Mode	Frequency (Hz)	Damping (%)	Description
1	4.35	0.200	1st Flap Bending
2	6.42	0.253	1st Lag Bending
3	11.5	0.243	2nd Flap Bending
4	20.5	0.283	3rd Flap Bending
5	23.1	0.310	2nd Lag Bending
6	35.3	0.307	4th Flap Bending
7	46.4	0.428	1st Torsion
8	48.6	0.422	3rd Lag Bending
9	54.6	0.455	5th Flap Bending

Los Alamos NATIONAL LABORATORY EST. 1943

NREL vs. LANL Fixed-Free



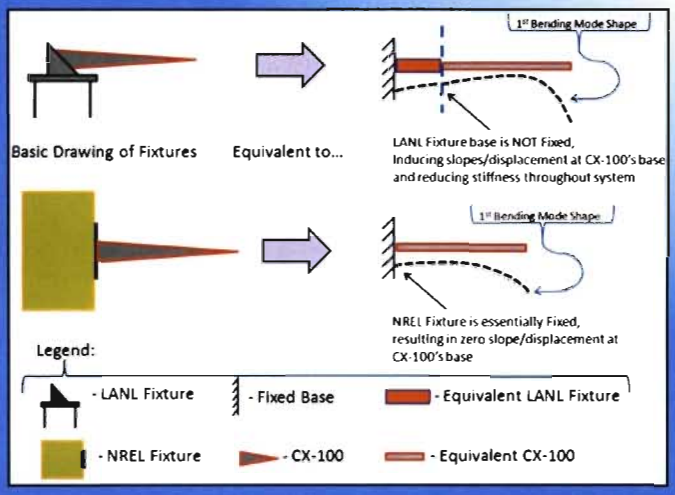
VS.



Mode	Description	NREL Fixed-Free Frequency (Hz)	LANL Fixed-Free Frequency (Hz)	% Difference
1	1st Flap Bending	4.35	3.37	26%
2	1st Lag Bending	6.42	3.85	53%
3	2nd Flap Bending	11.5	8.77	27%
4	3rd Flap Bending	20.5	16.8	20%
5	2nd Lag Bending	23.1	19.3	18%
6	4th Flap Bending	35.3	31.0	13%
7	1st Torsion	46.4	38.7	18%
8	3rd Lag Bending	48.6	45.7	6.2%
9	5th Flap Bending	54.6	52.3	4.3%

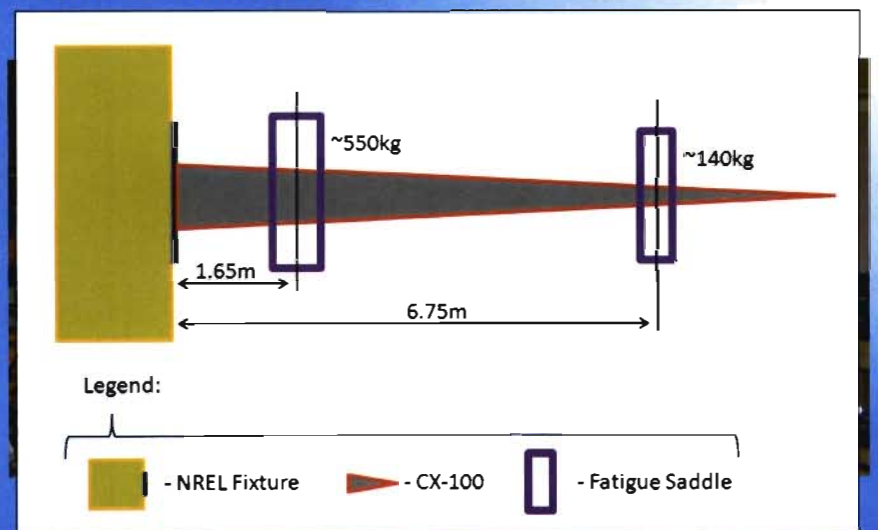
Los Alamos NATIONAL LABORATORY EST 1943

NREL vs. LANL Fixed-Free



Los Alamos NATIONAL LABORATORY EST 1943

NREL Fixed-Free with Fatigue Masses



Los Alamos NATIONAL LABORATORY EST. 1943

NREL Fixed-Free with Fatigue Masses



Mode	Frequency (Hz)	Damping (%)	Mode Description
1	1.82	0.132	1st Flap Bending
2	2.68	0.312	1st Lag Bending
3	9.23	0.546	2nd Flap Bending
4	12.7	0.868	3rd Flap Bending
5	14.7	1.50	2nd Lag Bending
6	18.9	0.958	4th Flap Bending
7	24.4	1.33	3rd Lag Bending

Los Alamos NATIONAL LABORATORY EST. 1943

Outline

- Background
- Objectives
- Modal Analysis
- Accelerometer vs. Piezoelectric Sensor
- Structural Health Monitoring
- Conclusions



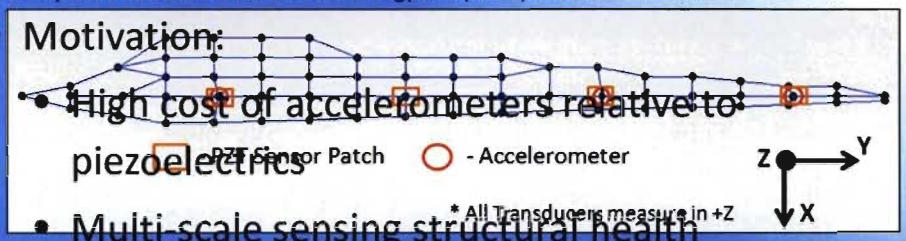


Los Alamos NATIONAL LABORATORY EST. 1943

Accelerometers vs. Piezoelectric Sensors for Modal Analysis

Completed at National Renewable Energy Lab (NREL) in Fixed-Free Condition

- Motivation:
  - High cost of accelerometers relative to piezoelectrics
  - Multi-scale sensing structural health monitoring approach



Mode	Accelerometer Data		PZT Sensor Data		% Difference	
	Frequency (Hz)	Damping (%)	Frequency (Hz)	Damping (%)	Frequency (Hz)	Damping (%)
1	4.35	0.170	4.35	0.176	0.00%	3.47%
2	6.42	0.255	6.42	0.257	0.00%	0.78%
3	11.5	0.260	11.5	0.254	0.00%	2.33%
4	20.5	0.284	20.5	0.278	0.00%	2.14%
5	23.1	0.324	23.1	0.318	0.00%	1.87%
6	35.3	0.313	35.3	0.302	0.00%	3.58%

Los Alamos NATIONAL LABORATORY EST. 1943

Outline

- Background
- Objectives
- Modal Analysis
- Accelerometer vs. PZT Sensor
- Structural Health Monitoring
- Conclusions





Los Alamos NATIONAL LABORATORY EST. 1943

Structural Health Monitoring (SHM)

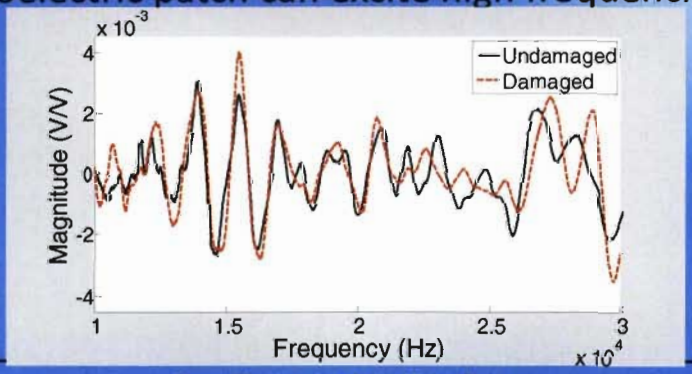
- Full-scale fatigue test of 9 m CX-100 blade at National Renewable Energy Lab (NREL)
- Goal: develop methods to detect and monitor damage progression
- How: high and low frequency sensing techniques
- Where: blade root and max chord

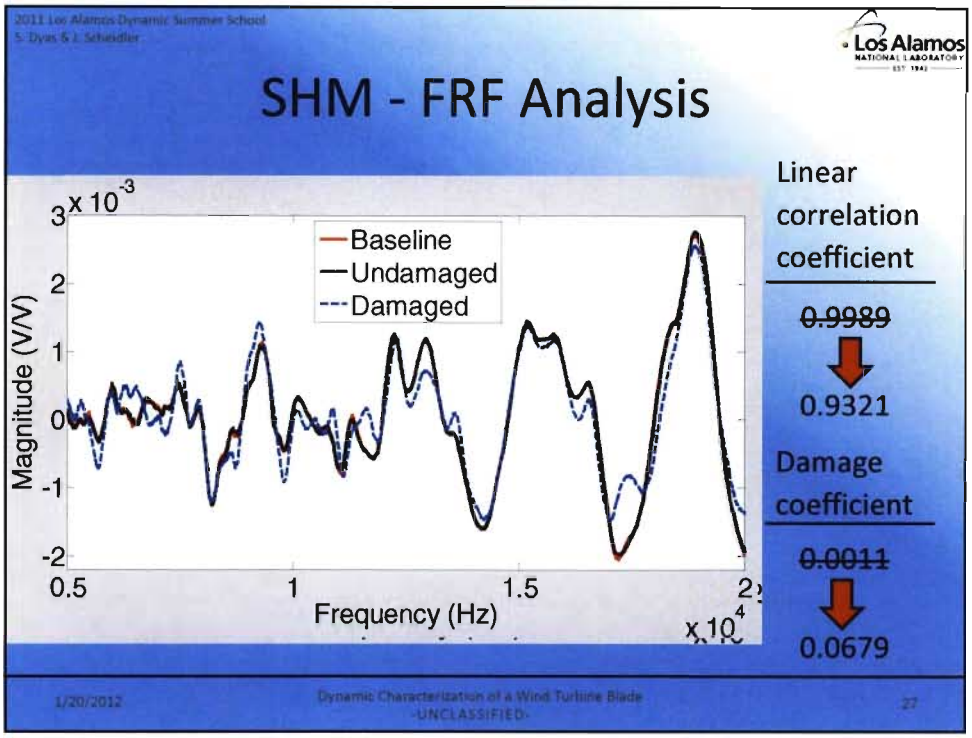


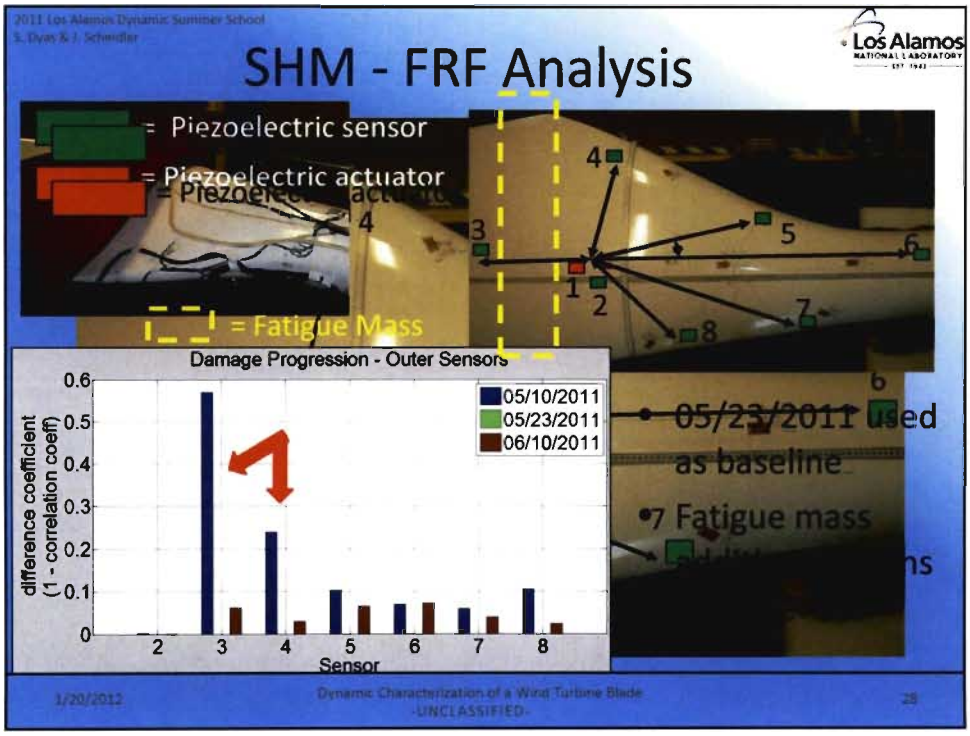
Los Alamos NATIONAL LABORATORY EST. 1943

SHM - Damage Detection via FRF Analysis

- Frequency response function (FRF) is a function of structure mass, stiffness, and damping
- Piezoelectric patch can excite high frequencies



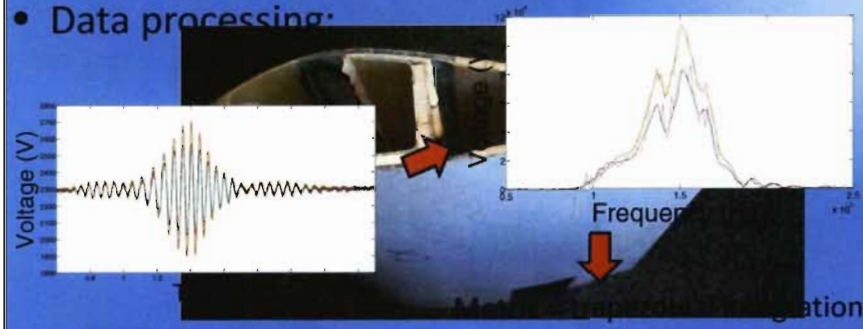

2011 Los Alamos Dynamic Summer School
S. Dyas & J. Schneider
Los Alamos NATIONAL LABORATORY
SHM - FRF Analysis
x 10^-3
Magnitude (V/V)
Baseline
Undamaged
Damaged
Frequency (Hz)
x 10^4
Linear correlation coefficient
0.9989
0.9321
Damage coefficient
0.0011
0.0679
1/20/2012
Dynamic Characterization of a Wind Turbine Blade
-UNCLASSIFIED-
27


2011 Los Alamos Dynamic Summer School
S. Dyas & J. Schneider
Los Alamos NATIONAL LABORATORY
SHM - FRF Analysis
= Piezoelectric sensor
= Piezoelectric actuator
= Fatigue Mass
Damage Progression - Outer Sensors
difference coefficient (1 - correlation coeff)
05/10/2011
05/23/2011
06/10/2011
Sensor
05/23/2011 used as baseline
7 Fatigue mass
1/20/2012
Dynamic Characterization of a Wind Turbine Blade
-UNCLASSIFIED-
28

Los Alamos NATIONAL LABORATORY EST. 1943

SHM – Lamb Wave Propagation Technique

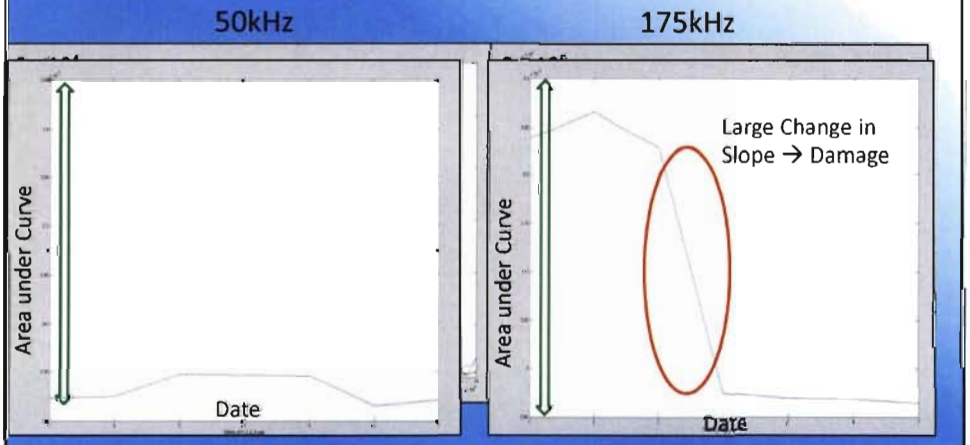
- 11 excitation frequencies between 50 and 275 kHz
- Complex composite structure
- Data processing:



ation of FFT
(energy content of response)

Los Alamos NATIONAL LABORATORY EST. 1943

SHM – Lamb Wave Propagation



Los Alamos NATIONAL LABORATORY EST. 1943

Outline

- Background
- Objectives
- Modal Analysis
- Accelerometer vs. Piezoelectric Sensor
- Structural Health Monitoring
- Conclusions

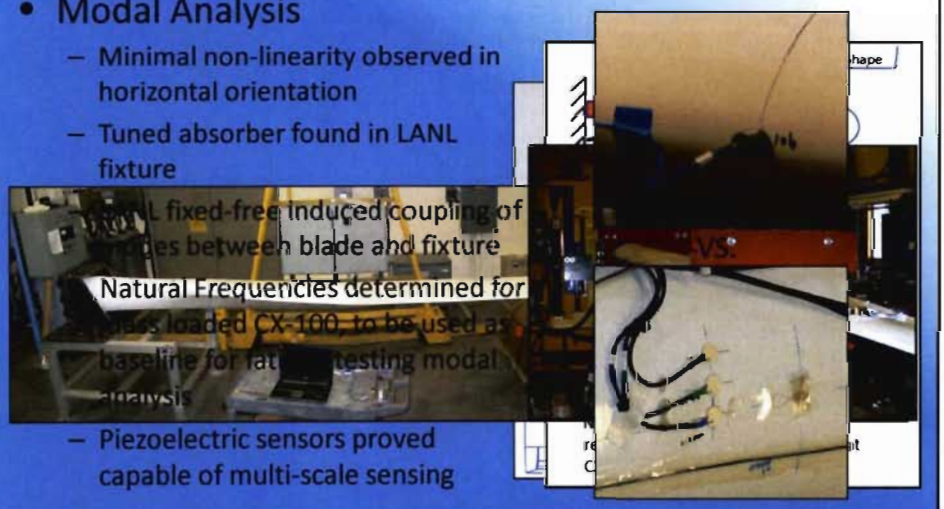




Los Alamos NATIONAL LABORATORY EST. 1943

Conclusions

- Modal Analysis
  - Minimal non-linearity observed in horizontal orientation
  - Tuned absorber found in LANL fixture
  - LANL fixed-free induced coupling of modes between blade and fixture
  - Natural Frequencies determined for glass loaded CX-100, to be used as baseline for fatigue testing modal analysis
  - Piezoelectric sensors proved capable of multi-scale sensing



Los Alamos NATIONAL LABORATORY EST. 1943

Conclusions

- Structural Health Monitoring
  - High-frequency FRF sensing technique insensitive to operating conditions
  - Multiple excitation frequencies required for high-frequency lamb wave propagation
  - Framework developed for multi-scale sensing SHM



Los Alamos NATIONAL LABORATORY EST. 1943

Acknowledgement

- Dr. Charles Farrar — Los Alamos NATIONAL LABORATORY
- Dr. Peter Avitabile — UMASS LOWELL
- Vibrant Technology Inc. — ME'scopeVES 5.0™ Visual Engineering Series
- SIMULIA — SIMULIA ABAQUS

Los Alamos NATIONAL LABORATORY EST 1943

Summary

- Modal Analysis
  - Minimal non-linearity observed in horizontal orientation
  - Tuned absorber found in LANL fixture
  - LANL fixed-free induced coupling of modes between blade and fixture
  - Natural Frequencies determined for mass loaded CX-100, to be used as baseline for fatigue testing modal analysis
  - Piezoelectric sensors proved capable of multi-scale sensing
- Structural Health Monitoring
  - High-frequency FRF technique insensitive to operating conditions
  - Multiple excitation frequencies required for high-frequency lamb wave propagation
  - Framework developed for multi-scale sensing SHM

Acknowledgement

Dr. Charles Farrar → Los Alamos

UMASS LOWELL ← Dr. Peter Avitabile

Vibrant Technology Inc. → ME'scopeVES 5.0 Visual Engineering Series

ABAQUS ← SIMULIA

Los Alamos NATIONAL LABORATORY EST 1943

SHM – Inner Sensor Array Reference

